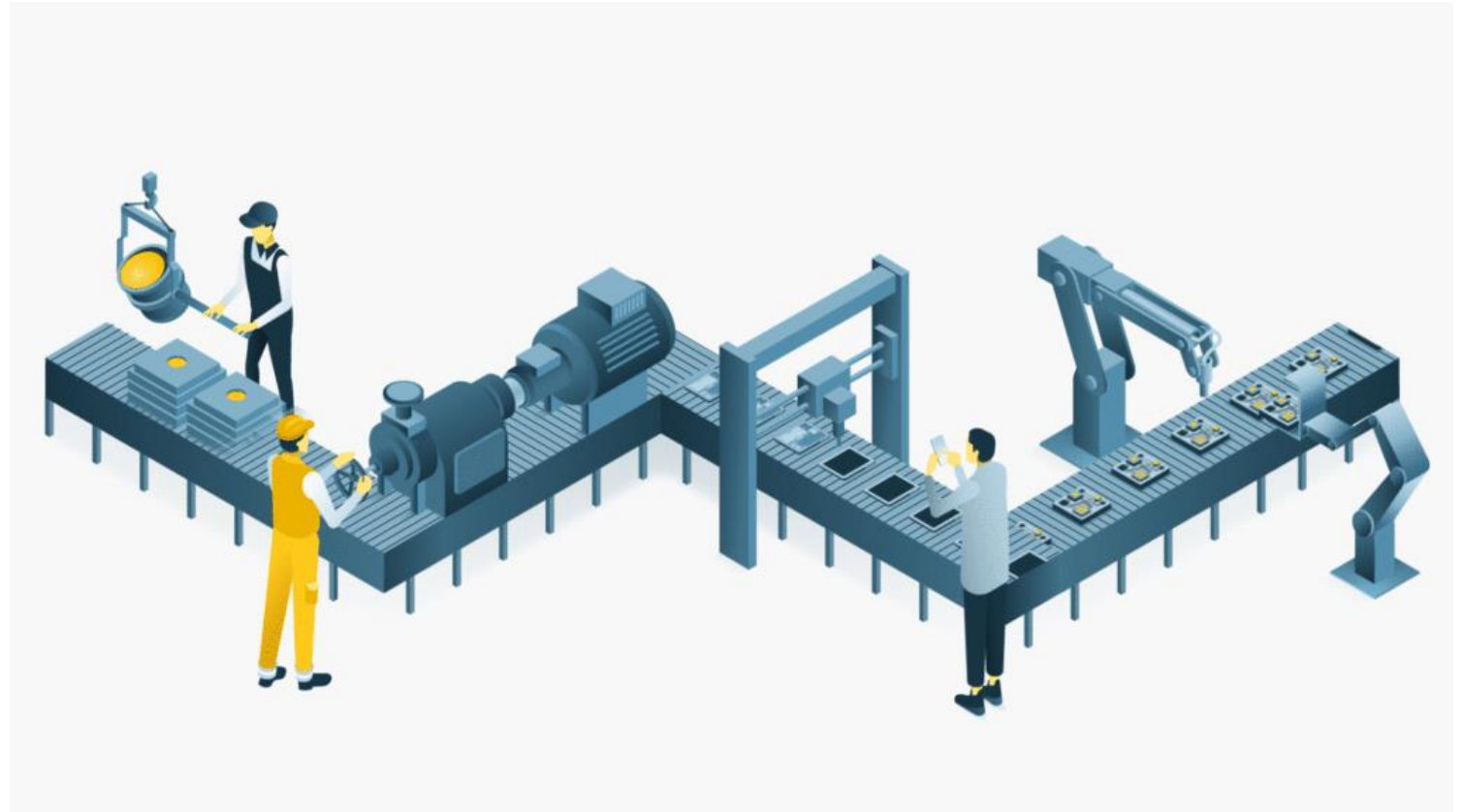

PartInspect

A SIMULATED ROS2 AND
GAZEBO ENVIRONMENT FOR
AUTOMATED QUALITY
INSPECTION IN
MANUFACTURING.



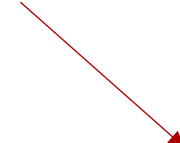


The ACTORS

- A CONVEYOR BELT
- A VISION SYSTEM (made of 2 cameras alongside the conveyor belt)
- A SOLENOID ACTUATOR (which can push objects away)
- OBJECTS (**cubes** and **cones**)

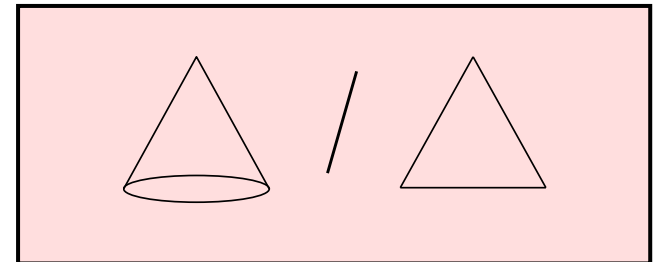
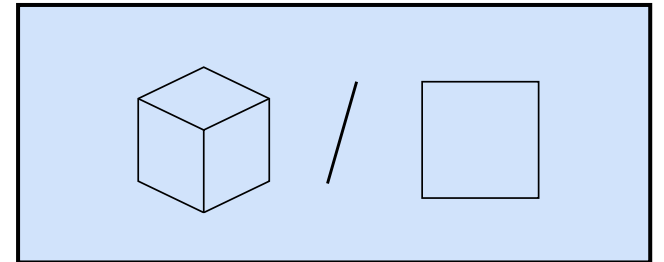
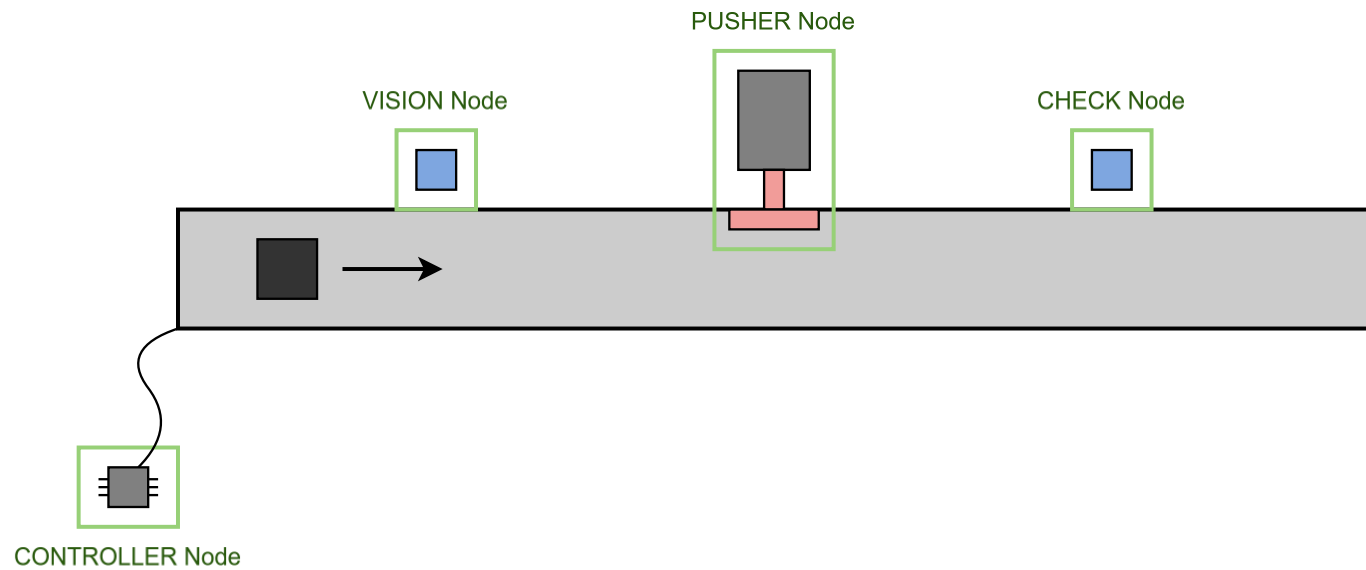


Good parts (preserved)

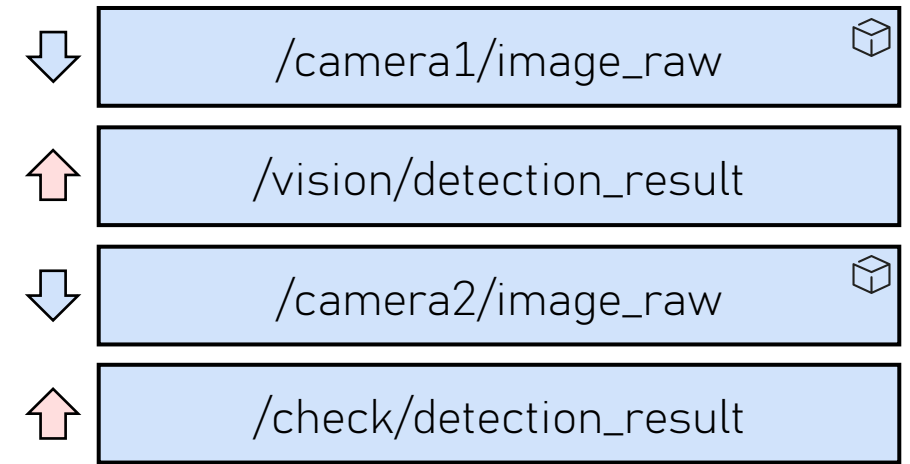


Defective parts (rejected)

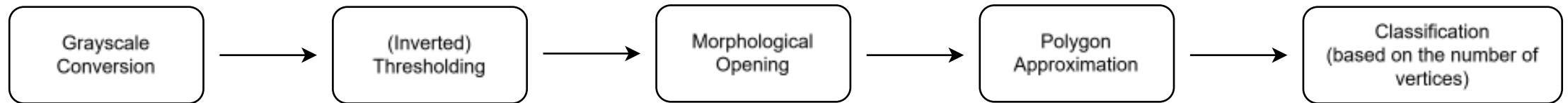
SYSTEM OVERVIEW



VISION (& CHECK) Node

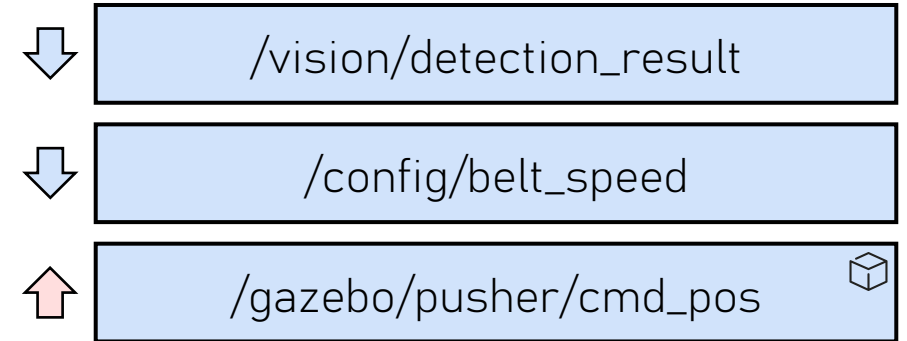


- VISION Node: detects defective parts and notifies the pusher
- CHECK Node: detects defective parts that were not pushed away and notifies the controller
- Both nodes receive images from simulated Gazebo cameras



The entire pipeline is applied to a specific **Region of Interest (ROI)**, to ensure that the object is fully visible in the image

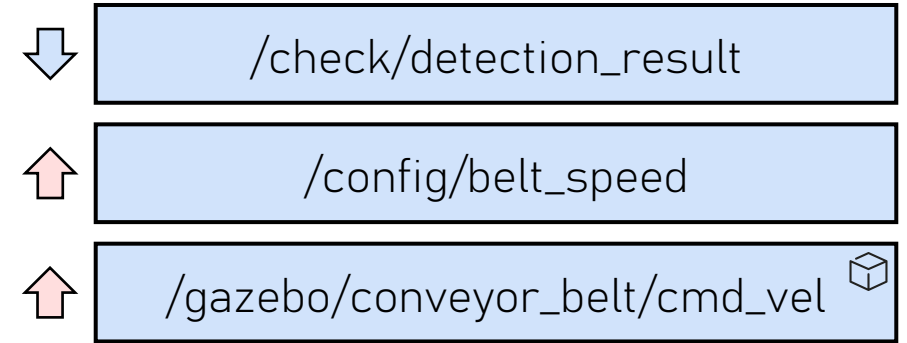
PUSHER Node



- TRACKS MOTION of defective parts (once detected from the vision node)
 - Actuation time is automatically computed based on the conveyor belt speed
 - If belt speed changes between detection and actuation, the actuation time changes accordingly
- Controls **EXTENSION** and **RETRACTION** of a simulated PID-controlled **SOLENOID**
 - Retraction happens 500ms after the extension

CONTROLLER Node

- CONVEYOR BELT controller
 - Supports variable belt speed values
 - Default belt speed is set 1 m/s
- Continuously publishes belt speed to pusher node and commands simulated conveyor belt
 - Continuous publishing ensures simulation stability and actuation accuracy
- Immediately stops if the check node detects a defective part



Other ROS NODES

- **CLEANER** Node: removes fallen object from the simulation
 - Invokes a Gazebo service to delete objects
 - Removal occurs 15 seconds after they reach the ground
- **DUMMY CAMERA** Node: generates fake camera images
 - Provides a mock interface for a Gazebo camera
 - Enables system testing without a Gazebo simulation



CONCURRENCY

